

altairsim — CP/M 2.2b on an Altair 8" floppy

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To launch it, name the machine file from inside this folder:

```
$ altairsim cpm22-buffered.toml
```

On **Windows** the program is `altairsim.exe`:

```
> altairsim.exe cpm22-buffered.toml
```

The machine boots itself — you type no `B00T` command — and comes up at the CP/M prompt. Type `DIR` to see the disk:

```
56K CP/M 2.2b v2.3  
For Altair 8" Floppy  
  
A>DIR
```

Mike Douglas's track-buffered **CP/M 2.2b v2.3**, on an 8" Pertec FD-400 behind an 88-DCDD, booted by the DBL PROM at `FF00` — `RUN FF00` is the machine file's whole startup, because on a real disk Altair that was `EXAMINE FF00` and `RUN`.

`^E` (STOP) takes the keyboard back to the monitor at any point; `RUN` resumes. `^C` belongs to CP/M (it is warm boot) and CP/M gets it.

The files

File	What it is
<code>cpm22-buffered.toml</code>	The machine: <code>base = "default"</code> plus the floppy in drive 0. Read it — it explains the memory arithmetic and what <code>readonly</code> really does on this controller.
<code>cpm22-terminal.toml</code>	The same machine and the same disk, but the console is a built-in VT100 window the simulator draws itself instead of stdio. The monitor stays in the terminal you launched from; CP/M comes up in its own window. No telnet client, no external emulator. Try <code>emulation=adm3a</code> for a period CP/M terminal.
<code>cpm22-fdcplus.toml</code>	CP/M from an FDC+ Serial Drive Server : no disk in this folder is used. The FDC+ gets its tracks from the server over a serial cable. Set your serial port in the file first — see below.
<code>cpm22-fdcplus-hdf.toml</code>	CP/M on the FDC+'s 1.5 MB floppy (drive type 5), from <code>CPM22-48K-HDF.dsk</code> . The stock DBL boot PROM boots it, although it cannot read the disk — see below.
<code>CPM22-48K-HDF.dsk</code>	The 1.5 MB CP/M 2.2b disk, for a 48K machine: <code>MOVCPM</code> , <code>COPY</code> , <code>PCGET</code> / <code>PCPUT</code> , <code>MBASIC</code> , <code>WM</code> , and games, with 920K free.
<code>cpm22b23-56k.dsk</code>	The bootable system disk, built for a 56K machine. Carries <code>DDT.COM</code> , <code>M80</code> / <code>L80</code> , <code>MBASIC</code> and the host-bridge utilities (<code>R</code> , <code>W</code> , <code>HDIR</code>), with 18K free. Shared by both machine files above.

There is no undo. Drive 0 is mounted read/write because that is what a real machine is, and CP/M writes to `A:` for anything you create. In a clone `git checkout` puts the image back; in the package you were handed, nothing does. Copy it first if you are about to test writes in anger.

Three drives are empty. `MOUNT dsk0:drive1 "my-scratch.dsk"` fills one, or `FORMAT` from inside CP/M will make you a disk.

`BOOT.ASM` and `BIOS.ASM` — this CP/M's own, and the authoritative source the 88-DCDD was built from — are in `disks/mits-88dcdd/cpm22/buffered/`. They are source rather than product, so they stay in the repository and are not in the package.

CP/M from an FDC+ drive server

`cpm22-fdcplus.toml` is the same Altair with an **FDC+** in place of the 88-DCDD, in its serial-drive mode. The disk images are on an **FDC+ Serial Drive Server** (or any program that speaks its protocol) at the other end of a serial cable, and the board gets them a track at a time. The simulator and a real FDC+ Altair can use the same images through the same server.

1. Start the server, and mount a bootable 8" CP/M image in its drive 0. `cpm22b23-56k.dsk` from this folder will do.

2. Open `cpm22-fdcplus.toml` and set `connect` on `fdc0` to your serial port. The file has examples for macOS (`serial:/dev/cu.usbserial-XXXX`), Linux (`serial:/dev/ttyUSB0`) and Windows (`serial:COM3`).
3. Set `baud` to the rate that the server uses: 9600, 19200, 38400, 57600, 76800, 230400, 403200 or 460800. The two ends must agree.
4. Run it:

```
$ altairsim cpm22-fdcplus.toml
```

If the port does not open, the error lists the serial ports this computer has.

The file sets `clock_hz = 10000000` and `baud = 230400`. The crystal must not be full speed: the CP/M BIOS gives up on a sector after a count that takes 0.28 seconds at 10 MHz, and a track takes 0.19 seconds at 230400 baud. At full speed the count takes a few milliseconds and no track arrives in time. On a slower line, slow the crystal down too: at 38400 baud a track takes about one second, which needs the real 2 MHz.

CP/M on the FDC+'s 1.5 MB floppy

`cpm22-fdcplus-hdf.toml` is the same Altair with an **FDC+** in place of the 88-DCDD, with its drive type switches at 5: a high-density floppy that holds 1.5 MB, one 10,240-byte sector to a track. `CPM22-48K-HDF.dsk` is in drive 0. Run it:

```
$ altairsim cpm22-fdcplus-hdf.toml
```

and you get:

```
48K CP/M 2.2b v1.2
For Altair 1.5Mb Floppy

A>
```

The boot PROM at `FF00` is the same DBL as always, and it cannot read this disk. The FDC+ gives it a sector of its own, from the board's memory, and that sector holds a loader that reads the real disk. The machine file explains it.

Leave the clock at its default. The CP/M for this disk moves each track at a speed that only a 2 MHz processor matches, and at full speed the board keeps 2 MHz time too. A faster `clock_hz` breaks it, as a faster processor would on a real Altair.